

Autoregressive Models: NADE

Suppose the dataset $D \subset \{0, 1\}^n$ are binary. Let $x \in D$ and write $x = (x_1, x_2, \dots, x_n)$ for $x_i \in \{0, 1\}$. We can write the likelihood $p(x)$ as

$$p(x) = \prod_{i=2}^n p(x_i | x_{<i}) p(x_1),$$

where $x_{<i} = (x_1, \dots, x_{i-1})$.

In an autoregressive model we assume the binary x_i 's follow Bernoulli distributions. Specifically, $x_1 \sim B(\theta_1)$ and, for $i > 1$,

$$p_{\theta_i}(x_i | x_{<i}) = B(f_{\theta_i}(x_{<i})),$$

where θ_i 's are trainable parameters and $f_{\theta_i} : \{0, 1\}^{i-1} \rightarrow [0, 1]$.

(1) In a **Fully-Visible Sigmoid Belief Network (FVSBN)**, each f_{θ_i} is just a linear layer:

$$f_{\theta_i}(x_{<i}) = \sigma \left(A_i \begin{pmatrix} x_1 \\ \vdots \\ x_{i-1} \end{pmatrix} \right),$$

where σ is the sigmoid function and $A_i \in \mathbb{R}^{1 \times (i-1)}$ contains the parameters. Total parameters $\sim O(n^2)$

(2) To improve the expressiveness of FVSBN, we may use MLP for each f_{θ_i} :

$$\begin{aligned} f_{\theta_i}(x_{<i}) &= \sigma(\alpha_i h_i + b_i) \\ h_i &= \sigma(A_i x_{<i} + c_i) \in \mathbb{R}^d, \end{aligned}$$

where $\alpha_i \in \mathbb{R}^{1 \times d}$, $b_i \in \mathbb{R}$, $A_i \in \mathbb{R}^{d \times i}$ and $c_i \in \mathbb{R}^d$. Total parameters $\sim O(n^2d)$.

(3) **Natural Autoregressive Density Estimator (NADE)** uses a shared weight matrix $W \in \mathbb{R}^{d \times n}$ for all the conditional probabilities:

$$\begin{aligned} f_{\theta_i}(x_{<i}) &= \sigma(\alpha_i h_i + b_i) \\ h_i &= \sigma(W_{\cdot, <i} \cdot x_{<i} + c). \end{aligned}$$

Total parameters $\sim O(nd)$.

To learn the model our objective function is the KL divergence:

$$\begin{aligned} \min_{\theta} \text{KL}(p_{data} \parallel p_{\theta}) &= \min_{\theta} \mathbb{E}_{x \sim p_{data}} [\log p_{data}(x) - \log p_{\theta}(x)] \\ &= \min_{\theta} \mathbb{E}_{x \sim p_{data}} [-\log p_{\theta}(x)] \\ &= \max_{\theta} \mathbb{E}_{x \sim p_{data}} [\log p_{\theta}(x)], \end{aligned}$$

which is also to max the likelihood.